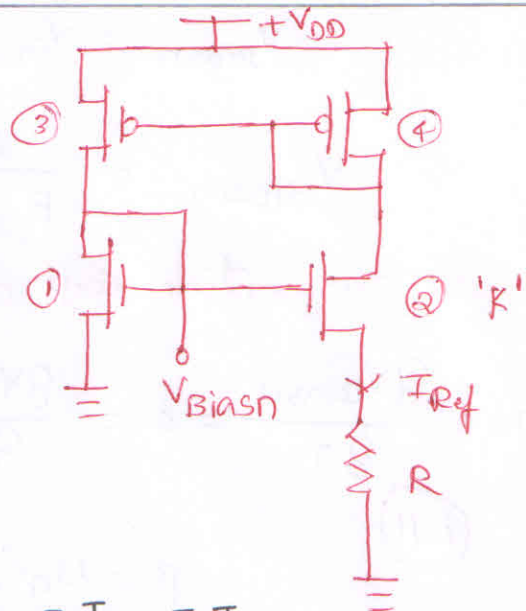


SOLUTION QUESTION NO. 81:-

1 (a) i) Derive an expression for I_{Ref} .

$$V_{GS1} = \sqrt{\frac{2I_{D1}}{\beta_1}} + V_{THN1} \rightarrow (1)$$

$$V_{GS2} = \sqrt{\frac{2I_{D2}}{\beta_2}} + V_{THN2} \rightarrow (2)$$



Given that:- $M3$ & $M4$ are matched

$$\Rightarrow I_{D3} = I_{D1} = I_{D4} = I_{D2} = I_{Ref}$$

Body effect is neglected ($\gamma=0$)

$$\therefore V_{THN2} = V_{THN1}$$

$$\beta_1 = \mu_n c'_{ox} \left(\frac{W}{L} \right)_1, \quad \beta_2 = \mu_n c'_{ox} \left(\frac{W}{L} \right)_2$$

$M2$ is ' k ' times wider than $M1$, $(W/L)_2 = k(W/L)_1$

$$V_{GS1} = V_{GS2} + I_{Ref} \cdot R \rightarrow (1)$$

$$\sqrt{\frac{2I_{Ref}}{\beta_1}} - \sqrt{\frac{2I_{Ref}}{\beta_2}} = I_{Ref} \cdot R$$

$$\sqrt{\frac{2}{\beta_1}} \sqrt{I_{Ref}} \left[1 - \frac{1}{\sqrt{k}} \right] = I_{Ref} \cdot R$$

Two solutions exist $I_{Ref} = 0$ not desirable.

$$I_{Ref} = \frac{2}{R^2 \beta} \left[1 - \frac{1}{\sqrt{k}} \right]^2 \rightarrow (2)$$

where $\beta_1 = \mu_n c'_{ox} \cdot \left(\frac{W}{L} \right)_1$

(ii) Expression for Reference voltage " V_{Bias} "

$$V_{Bias} = V_{GS1} = I_{Rq} \cdot R + V_{THN}$$

$$V_{Bias} = \frac{2}{R \cdot \beta} \left[1 - \frac{1}{\sqrt{k}} \right]^2 + V_{THN} \quad (2M)$$

(iii) Zero Temperature Coeff at $T=300K$ for Bias voltage.

$$\text{ie. } \frac{\partial V_{Bias}}{\partial T} = 0 = \frac{\partial V_{THN}}{\partial T} + \frac{1}{2} \left[-\frac{1}{R\beta} \frac{\partial R}{\partial T} - \frac{1}{\beta R} \frac{\partial \beta}{\partial T} \right] (k=4)$$

(2M)

$$\beta = \mu_n c'_{ox} \cdot (W/L)$$

c'_{ox} , W/L independent of Temperature (given)

$$\frac{1}{\beta} \frac{\partial \beta}{\partial T} \Rightarrow \frac{1}{\mu_n} \cdot \frac{\partial \mu_n}{\partial T} = \frac{-1.5}{300(K)}$$

$$-\frac{\partial V_{THN}}{\partial T} = -\frac{1}{R\beta} \left[\frac{1}{R} \frac{\partial R}{\partial T} + \frac{1}{\beta} \frac{\partial \beta}{\partial T} \right]$$

$$-(-0.6mV) = -\frac{1}{R\beta} \left[2000 \times 10^{-6} - \frac{1.5}{300} \right]$$

$$1.2 \times 10^{-3} = -\frac{1}{R\beta} \left[-3 \times 10^{-3} \right]$$

$$\frac{R\beta}{2} = \frac{3}{1.2} = 2.5$$

$$\frac{R}{2} = \frac{2.5}{\beta} = \frac{2.5}{140 \times 10^{-6} \times 5}$$

$$\frac{R}{2} = 3.5714 \text{ K}\Omega \times 2 \rightarrow (5M)$$

$$\boxed{R \approx 7.14 \text{ K}\Omega}$$

5b) Given Data:- $V_{DD} = 3.3V$

PMOS M2
 $V_{THP1} = 0.8V$

NMOS M1
 $V_{THN} = 0.7$

$$\left(\frac{W}{L}\right)_2 = 78.75$$

$$\left(\frac{W}{L}\right)_1 = 22.5$$

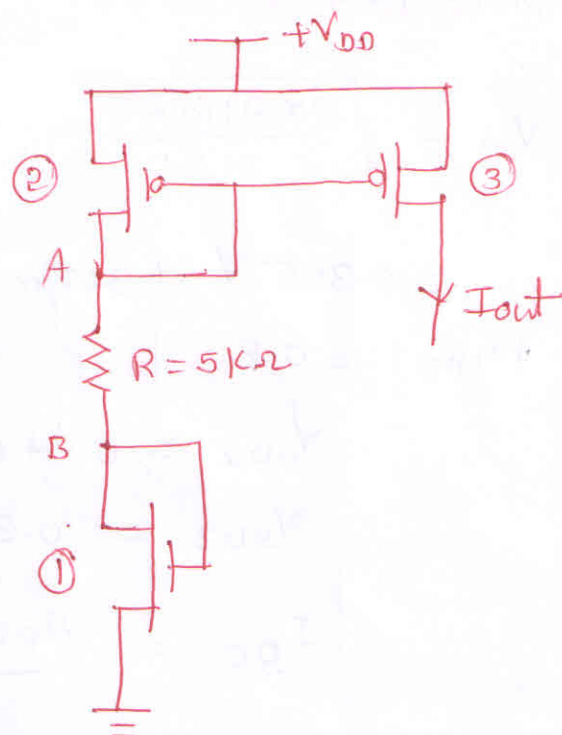
$$\mu_{pC'ox} = 40 \mu A/V^2$$

$$\mu_{nC'ox} = 140 \mu A/V^2$$

$$\beta_2 = 3150 \mu A/V^2$$

$$\beta_1 = 3150 \mu A/V^2$$

$$\beta_1 = \beta_2 = \beta$$



i) Current through $5k\Omega$ resistor:-

$$(I_{D1} = I_{D2}) I_R = \frac{V_A - V_B}{R} \quad (1M)$$

$$V_A = V_{DD} - V_{SG2}$$

$$V_B = V_{GS1}$$

$$V_A - V_B = V_{DD} - |V_{THP1}| - V_{OV2} - V_{THN} - V_{OV1}$$

$$= V_{DD} - |V_{THP1}| - V_{THN} - 2 \sqrt{\frac{2 \cdot I_R}{\beta}}$$

$$R \cdot I_R = 3.3 - 0.8 - 0.7 - 2 \sqrt{\frac{2 I_R}{\beta}}$$

$$5000 \times I_R = 1.8 - \frac{2\sqrt{2}}{\sqrt{3150 \times 10^{-6}}} \times \sqrt{I_R} \quad (2M)$$

$$-50.395 \sqrt{I_R} - 5000 I_R + 1.8 = 0$$

$$\sqrt{I_R} = 0.0145 (\checkmark) \quad -0.0246 (X)$$

$$I_R = 210.25 \mu A \rightarrow (2M)$$

ii) I_{out} needed = $0.5mA = 500\mu A$. ($\lambda = 0$)

$$W_3 = \left(\frac{500}{210.25}\right) W_2 = 2.378 W_2$$

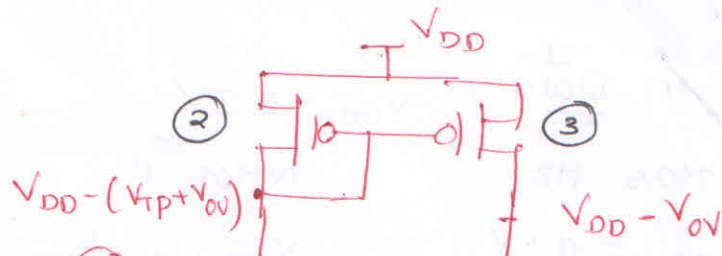
$$W_3 = 187.27 \mu m \rightarrow (2M)$$

iii) W_3 when $(\lambda \neq 0)$.

$$V_{OV} = \sqrt{\frac{2 \times 210.25}{3150}}$$

$$= 0.365 \text{ V } (365 \text{ mV})$$

$$|V_{THP}| = 0.8$$



(1M)

$$(\lambda_p = 0.1 \text{ V}^{-1})$$

$$V_{SD2} = 0.8 + 0.365 = 1.165 \text{ V}$$

$$V_{SD3} = 0.365 \text{ V.}$$

(1M)

$$I_{SD} = \frac{\mu_p C_{ox} (\frac{W}{L})}{2} [V_{SG} - |V_{TP}|]^2 (1 + \lambda V_{SD})$$

$$\frac{500}{210.25} = \frac{W_3}{W_2} \times \frac{[1 + 0.1(0.365)]}{[1 + 0.1(1.165)]}$$

$$W_3 = 1.077 \times \frac{500}{210.25} \times W_2$$

$$W_3 = 2.5611 \times W_2$$

$$W_3 = 201.73 \mu\text{m} \rightarrow (1M)$$

iv) % change in output current when V_{DD} changes by 5%.

$$\Delta I_R = \frac{\Delta V_{DD}}{\frac{1}{g_{m2}} + R + \frac{1}{g_{m1}}}$$

(1M)

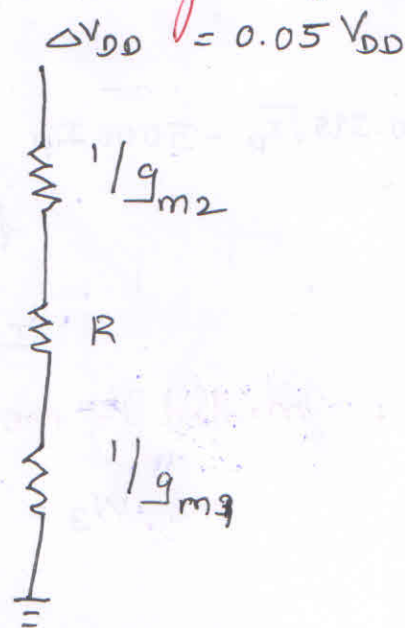
$$= \frac{0.05 \times 3.3}{5\text{K} + \frac{\sqrt{2} \times 10^4}{\sqrt{3150 \times 210.25}}}$$

$$\Delta I_R \approx 33 \mu\text{A} \quad 24.48 \mu\text{A}$$

$$\Delta I_{out} = 78.44 \mu\text{A} \quad 58.23 \mu\text{A}$$

$$(0.8) \quad 84.516 \mu\text{A} \quad 62.71 \mu\text{A}.$$

(1M)



Compre
Solution
Analog Sec. A

Q2 $V_{ov} = 0.2V$ Given

(i) Differential Voltage Gain $\frac{V_{out}}{V_{in}}$

$$\frac{V_{out}}{V_{in}} = |g_{m1} \times g_{m7} r_{o7} r_{o9} // g_{m3} r_{o3} (r_{o2} // r_{o5})| \quad (2)$$

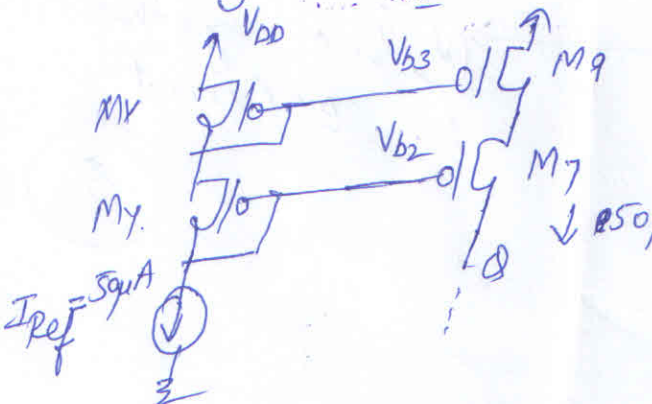
(ii) Resistance at node X = $(r_{o2} // r_{o5}) // \left(\frac{g_{m7} r_{o7} r_{o9} + r_{o3}}{1 + g_{m3} r_{o3}} \right) \quad (2)$

(iii) OCMR $\rightarrow$ Max = $V_{DD} - V_{ov9} - V_{ov7} = 3.3 - 0.2 - 0.2 = 2.9V \quad (2)$
Min = $V_{ov3} + V_{ov5} = 0.4V$

(iv) ICMR $\rightarrow$ Max = $V_{DD} - V_{ov_{ISS}} - V_{SG_{M1}} = 3.3 - 0.2 - 0.9V = 2.2V \quad (2)$
Min = $0.0V = V_{SS} = V_{Pmin} - 0.9V \approx 0 (Gnd) \quad (2)$

(v) DC voltage $V_{b2} = V_{DD} - V_{ov9} - V_{SG_{M7}} = 3.3 - 0.2 - V_{SG_{M7}} (= 1V) = 2.1V \quad (2)$

(vi)



Assuming Matching pairs

$$I_{Mx} = \mu_p C_{ox} \left(\frac{W}{L} \right)_x (V_{GS} - V_T)^2 \quad (1+1) = (2)$$

$$500\mu A = 40\mu A/V^2 \left(\frac{W}{L} \right)_x (V_{ov})^2$$

$$\left(\frac{W}{L} \right)_x = \frac{5}{4(0.2)^2} = \frac{5}{4 \times 0.04} = 6.3$$

$$\left(\frac{W}{L} \right)_y = \left(\frac{W}{L} \right)_x$$

(vii) $\omega_x = \frac{1}{\left[(r_{o5} // r_{o2}) // \left(\frac{g_{m7} r_{o7} r_{o9} + r_{o3}}{1 + g_{m3} r_{o3}} \right) \right] C_L} \quad \omega_R = \frac{1}{\left[g_{m10} r_{o10} r_{o8} // g_{m4} r_{o4} (r_{o2} // r_{o5}) \right] C_L}$

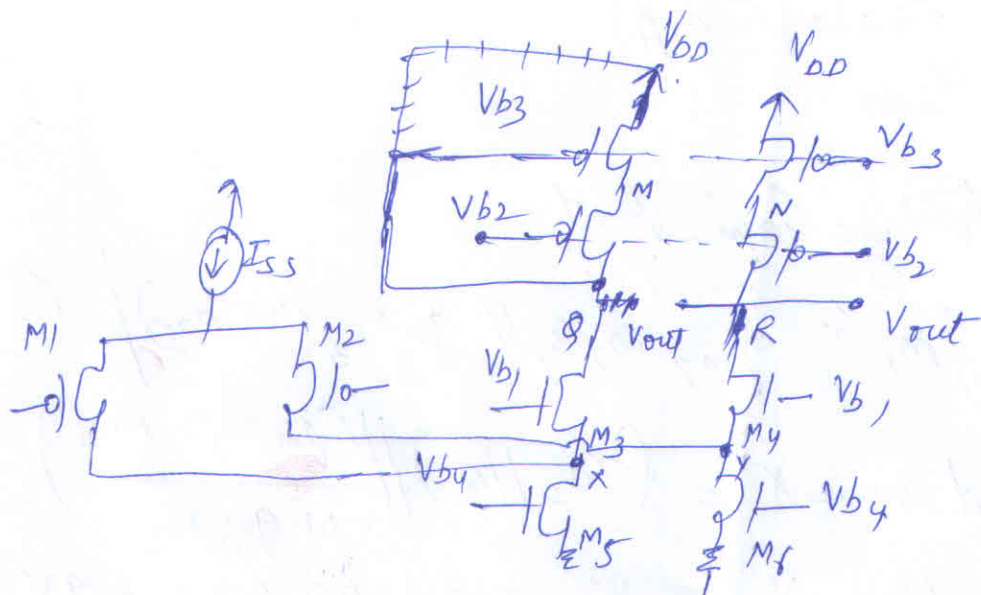
$$\omega_M = \frac{1}{\left[r_{o9} // \left(\frac{g_{m3} r_{o3} (r_{o2} // r_{o5}) + r_{o7}}{1 + g_{m7} r_{o7}} \right) \right] C_M}$$

$(3 \times 2) = 6$

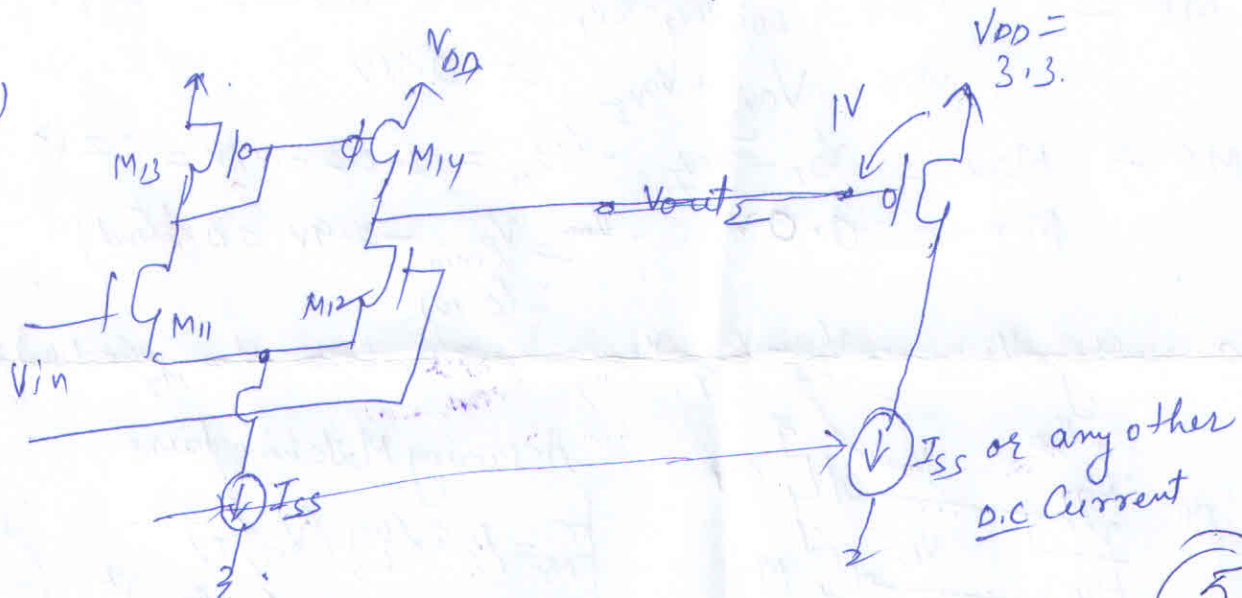
(viii)

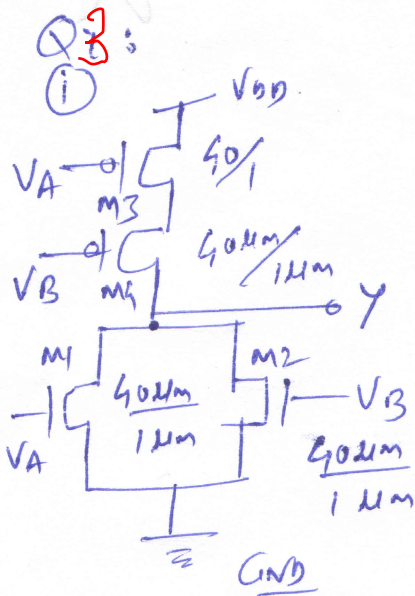
(P.T.O)

(Viii) Modified fig. 1.



(ix)





(ii)

Diffusion area at the o/p node

$$\begin{aligned}
 A_o &= (40 \mu\text{m} \times 10 \mu\text{m}) + (40 \mu\text{m} \times 20 \mu\text{m}) \\
 &= (40 \mu\text{m} \times 5 \mu\text{m}) + (40 \mu\text{m} \times 10 \mu\text{m}) \\
 &= 600 \mu\text{m}^2 = 600 \times 10^{-12} \text{ m}^2
 \end{aligned}$$

Total perimeter at o/p node

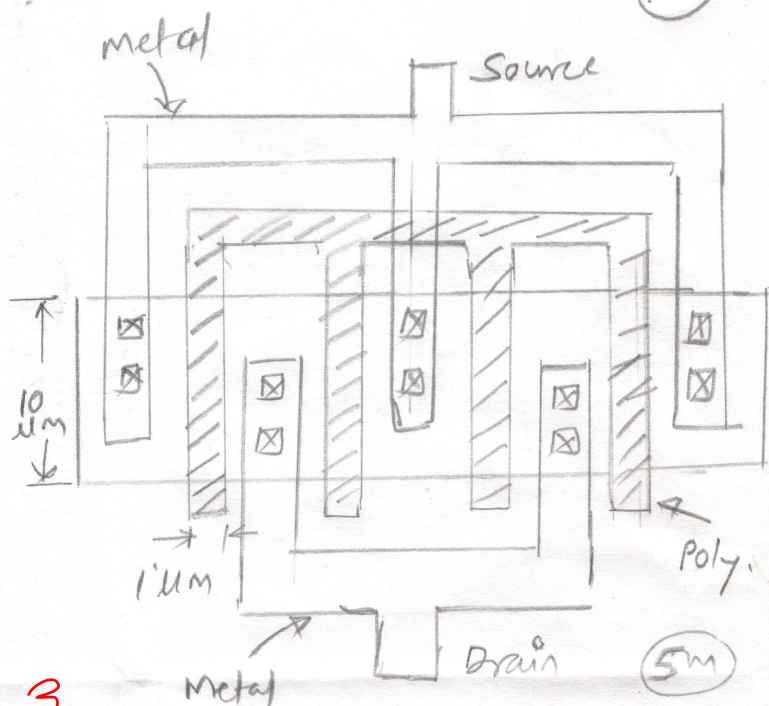
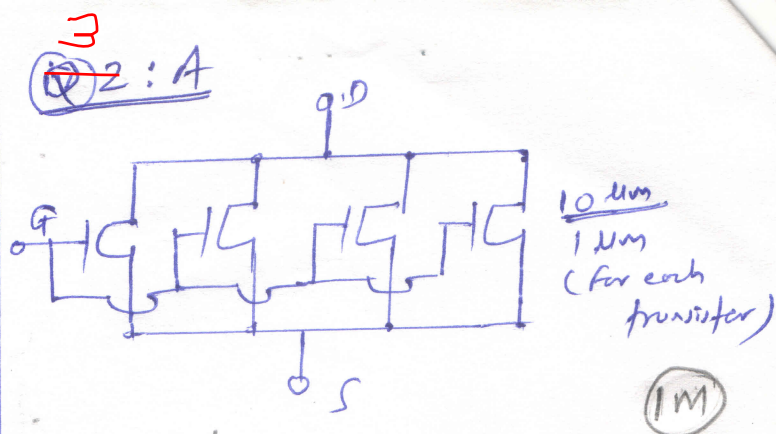
$$\begin{aligned}
 P_o &= [(2 \times 5 \mu\text{m}) + 40 \mu\text{m}] \\
 &\quad + [2 \times 10 \mu\text{m}] = 70 \mu\text{m} \\
 &= 70 \times 10^{-6} \text{ m}
 \end{aligned}$$

Total diffusion capacitance at o/p node

$$\begin{aligned}
 &= A_o C_{jo} + P_o C_{jsw} \\
 &= (600 \times 10^{-12} \times 5 \times 10^{-4}) \\
 &\quad + (70 \times 10^{-6} \times 5 \times 10^{-10}) \\
 &= 300 \text{ ff} + 35 \text{ ff} \\
 &= 335 \text{ ff}
 \end{aligned}$$

(iii) Total input cap. at node VA'

$$\begin{aligned}
 &= 2 \times W \times L \times C_{ox} \\
 &= 2 \times 40 \mu\text{m} \times 1 \mu\text{m} \times 5 \times 10^{-3} \\
 &= 2 \times 40 \times 1 \times 10^{-12} \times 5 \times 10^{-3} \\
 &= 400 \text{ ff}
 \end{aligned}$$



3

Q2: B

$$S = \frac{4 \mu\text{m}}{1 \mu\text{m}} = 4$$

$$f_{PHL} = \frac{C_L}{S} \frac{V_{DD}}{S E_{ox}} = \frac{I_{PHL}}{S^2}$$

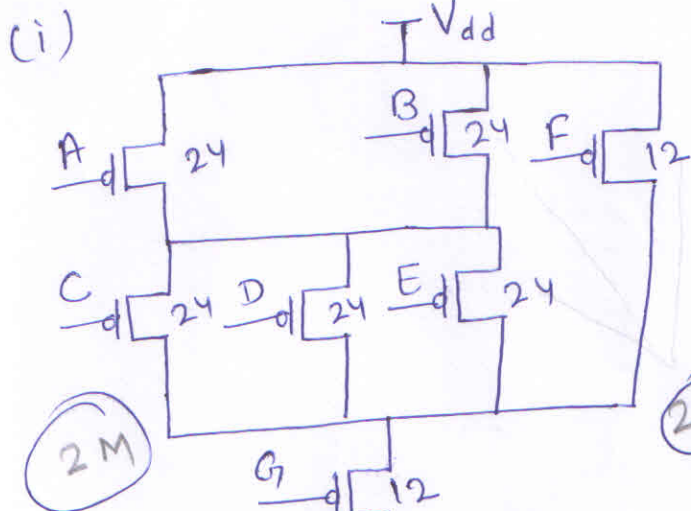
$$\begin{aligned}
 f_1 &= S^2 f = (4)^2 \times 100 \text{ MHz} \\
 &= 1.6 \text{ GHz}
 \end{aligned}$$

$$P' = S \cdot P = 4 \times 5 = 20 \text{ W}$$

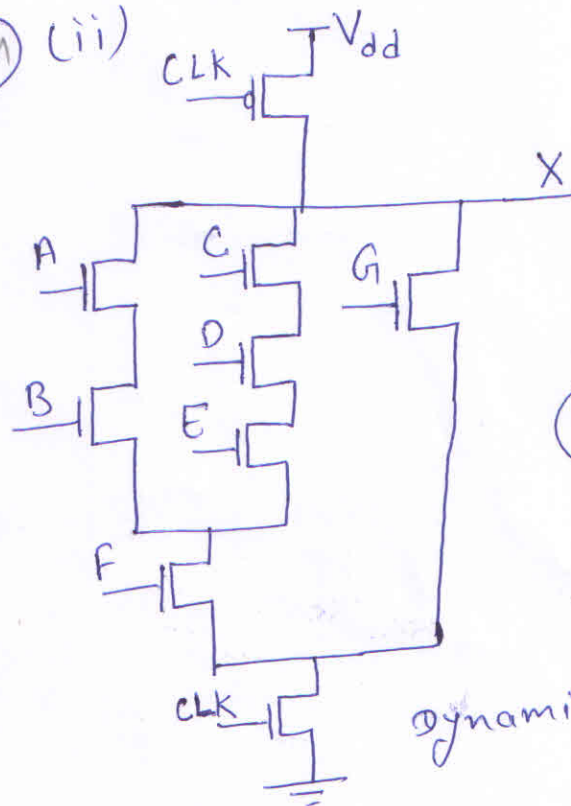
Sol:-

Q1(a) $X = ((\bar{A} + \bar{B})(\bar{C} + \bar{D} + \bar{E}) + \bar{F})\bar{G}$

$X = \overline{(AB + CDE)F + G}$



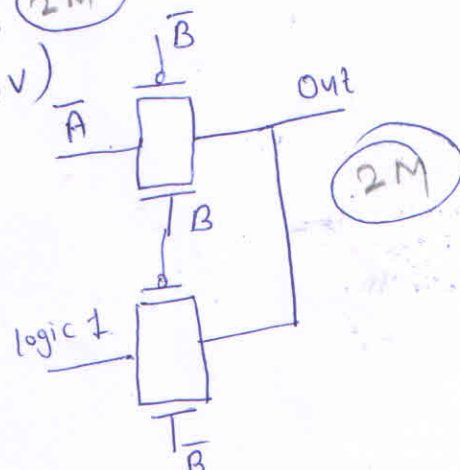
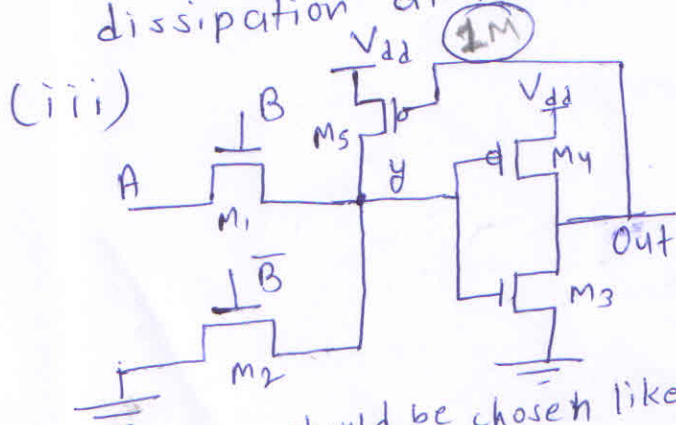
Input vectors for worse case pull-down resistance:-
 $ABCDEFGH = 0000001, 0011110$
 Best case pull down resistance when
 $ABCDEFGH = 1111111$



static CMOS

dynamic CMOS

Q2(b) (i) Circuit is NAND Gate, (ii) when $A=B=1$, voltage at node y is $V_y = V_{dd} - V_{tn}$. This causes static power dissipation at the inverter.



size of M_5 should be chosen like that when $A=B=0$, either M_1 or M_2 would be able to pull node y to $V_{dd}/2$ or less.

Solution:

$$a) \frac{101}{103} = \frac{(V_{DD} - V_A - V_{tN})^2}{2 \left[(V_{DD} - V_{tN}) V_A - \frac{V_A^2}{2} \right]}$$

$$2 = \frac{(2.1 - V_A)^2}{2 \left[2.1 V_A - \frac{V_A^2}{2} \right]} = \frac{(2.1 - V_A)^2}{4.2 V_A - V_A^2}$$

$$8.4 V_A - 2 V_A^2 = 4.41 - 4.2 V_A + V_A^2$$

$$3 V_A^2 - 12.6 V_A + 4.41 = 0$$

$$(5H) \quad \underline{V_A = 0.39} \quad \text{OR} \quad (3.81) \quad X.$$

$$b) \quad C = 512 \times 2 \times 6 \text{ fF}/\mu\text{m}^2 \times 0.5 \times 0.25$$

$$\underline{C = 768 \text{ fF}} \quad (5H)$$

Solution:

(1)

$$a) \tau_{clk} \geq t_{proc} + D + t_{setup} = \underline{250} + k(\underline{50}) + 3$$

$$1250 \geq 550 + k(50)$$

$$(3M) \quad k \leq 14 \quad (\text{gate delays})$$

$$b) \left. \begin{aligned} t_{pcq} &= t_{pcq,old} + 10ps = 300ps \\ t_{ccq} &= t_{ccq,old} + 50ps = 150ps \end{aligned} \right\} (1M)$$

$$\left. \begin{aligned} t_{setup} &= t_{setup,old} - 50 = 250ps \\ t_{hold} &= t_{hold,old} + 50 = 250ps \end{aligned} \right\} (2M)$$